

Praween Siritanasak

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📍 Chiangmai, Thailand

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Education

Ph.D. University of California, San Diego , Physics — Experimental Cosmology - Dissertation: "Precise Measurement of B-mode Polarization Signal from the Cosmic Microwave Background with the POLARBEAR and the Simons Array" - Awarded Royal Thai Government Scholarship (2010–2017)	San Diego, CA, USA 2009 – 2018
M.Sc. University of California, San Diego , Physics	San Diego, CA, USA 2009 – 2012
B.Sc. Mahidol University , Physics - First-class Honours - Sritrangthong Scholarship (2005–2009)	Bangkok, Thailand 2005 – 2009

Experience

National Astronomical Research Institute of Thailand (NARIT) , Researcher - CMB polarization instrumentation and detector development - Member of the POLARBEAR/Simons Array collaboration - Dark photon search: member of the Siam Dark Photon (SDP) Consortium	Chiangmai, Thailand Oct 2020 – present 6 years 1 month
Center for Astrophysics and Space Sciences (CASS), UC San Diego , Postdoctoral Research Fellow - Development of next-generation CMB receiver systems under Professor Brian Keating - Testing and characterization of transition-edge sensor (TES) bolometers	San Diego, CA, USA June 2018 – Oct 2020 2 years 5 months
Department of Physics, UC San Diego , Graduate Research Assistant The POLARBEAR and the Simons Array CMB polarization experiment	San Diego, CA, USA 2009 – 2018 9 years

Awards

Best Oral Presentation Award For "Siam-DP: Probing the Dark Sector with a High-Sensitivity 230 GHz SIS Receiver"	21st Siam Physics Congress 2026
Royal Thai Government Scholarship Full-ride graduate scholarship for doctoral study abroad (2010–2017)	Royal Thai Government 2010
Sritrangthong Scholarship Merit-based undergraduate scholarship (2005–2009)	Mahidol University 2005

Skills

CMB Instrumentation: POLARBEAR, Simons Array, TES detectors, receiver development, cryogenics, anti-reflection coatings

Data Analysis: CMB polarization analysis, B-mode detection, gravitational lensing, power spectrum estimation

Programming & Tools: Python, C, MATLAB, Mathematica, LaTeX, Solidworks, HFSS

Operating Systems: macOS, Linux (RedHat, Ubuntu, Linux Mint)

Languages

Thai
Native

English

Fluent

Presentations

- **Oral:** "Siam-DP: Probing the Dark Sector with a High-Sensitivity 230 GHz SIS Receiver," 21st Siam Physics Congress, Hua Hin, Thailand (2026) — *Best Oral Presentation Award*
- **Oral:** "The Current Status of POLARBEAR and Simons Array Experiment," 19th Siam Physics Congress, Ayutthaya, Thailand (2024)
- **Oral:** "Instrument Development of the CMB Polarization POLARBEAR-2 Experiment," 230th American Astronomical Society Meeting, Austin, Texas, USA (2017)
- **Poster:** "The Broadband Anti-Reflection Coated Extended Hemispherical Silicon Lenses for POLARBEAR-2 Experiment," 16th International Workshop on Low Temperature Detectors, Grenoble, France (2015)